

Validity & Measurement Error

Tutorial 5

Today's Focus

- Understanding **validity**: Does our measure capture what we intend?
- Understanding **measurement error**: How do our measurements deviate from truth?
- Applying these concepts to **real-world crime data**
- Evaluating research design choices

Definition: The extent to which the variable/measure corresponds to the concept it's supposed to measure.

For a measure to have validity:

- Should capture the concept as **fully** as possible
- Should **not** capture other concepts

Lack of validity: Variable systematically fails to capture the concept, and fails repeatedly in the same way. There is a story about why what we observe is due to something *other than* the concept.

Key test: If problems persist even when we imagine our measurement is flawless (numbers/labels are perfect), it is a **validity issue**.

Example: Validity Problems

Concept: “Deadliness of crime”

Invalid measure

Using **total violent crimes** to measure deadliness

- Most violent crimes are non-lethal (assault, robbery)
- Variable does not match concept
- This is a **validity failure**

Valid measure

Using **homicide rates** to measure deadliness

- Directly captures the concept of deaths
- Good concept-variable alignment

Measurement Error: From Measure to Variable

Two types of measurement error:

Systematic measurement error (Bias)

Error produced when our measurement procedure obtains values that are, **on average**, either too high, too low, or wrong.

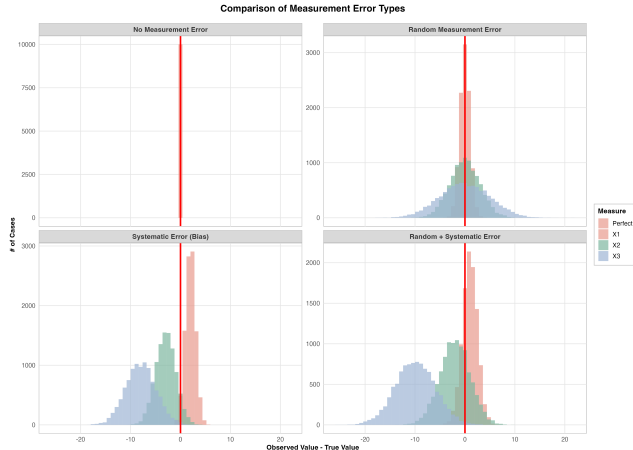
- Consistent direction of error
- Example: A scale that always adds 2 kg

Random (non-systematic) measurement error

Error derives from random features of the measurement process; produces observed values that are **sometimes too high and equally often too low**.

- No consistent direction
- Example: Minor fluctuations in scale readings

Visualizing Measurement Error Types



- **No error:** All observations = true value (perfect measurement)
- **Random error:** Spread around truth, but unbiased (mean ≈ 0)
- **Systematic error:** Consistently off target (mean $\neq 0$)

The Target Analogy

Accuracy vs. Precision:

- **High bias, low random error**
Precise but inaccurate (consistently off-target)
- **Low bias, high random error**
Accurate on average but imprecise (scattered)
- **Low bias, low random error**
Both accurate and precise (ideal!)

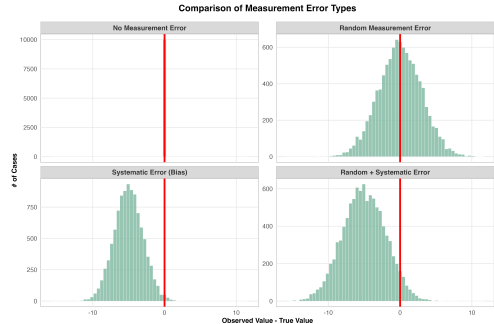


Figure: *

Red line = true value; distributions show measurement outcomes

Applying Concepts: Crime Data Case Study

We'll examine two documents:

- ① **The Dispatch editorial:** “Why the Perception That Crime Is Rising Persists”
 - Claims crime is rising under Biden administration
- ② **Jeff Asher's critique:** “Another Cautionary Crime Data Tale”
 - Points out measurement problems in the Dispatch piece

Your task: Identify validity and measurement error issues in both articles

Activity 1: Analyzing The Dispatch Editorial

Discuss with your group:

- ① What problems does the Dispatch identify with Biden administration's claims about crime?
- ② Are these problems due to:
 - Lack of **validity**? (wrong concept-variable match)
 - **Measurement error**? (problems in data collection/reporting)
- ③ If measurement error exists:
 - What **type**? (systematic bias or random error)
 - How might this affect conclusions about "crime is up"?
 - Could it lead to **wrongly accepting** or **wrongly rejecting** the claim?

Time: 10 minutes

Activity 2: Analyzing Asher's Critique

Now examine Asher's response:

- ① What problems does Asher identify with *The Dispatch* author's claims?
- ② Are these problems due to:
 - Lack of **validity**?
 - **Measurement error**?
- ③ If measurement error exists:
 - What **type**?
 - How might this affect conclusions about "crime is up"?
 - Direction of bias? (toward accepting or rejecting the claim)

Time: 10 minutes

Key Issues in the Crime Data Debate

Major problems identified by Asher:

① Comparing incomparable data sources

- NIBRS coverage: 10% (1998) vs. 75% (2022)
- Systematic bias: expansion looks like crime increase

② Using raw totals instead of rates

- Ignores 70M population growth
- Systematic bias toward “crime is up”

③ Validity failure: conflating concepts

- Using “violent crime totals” to claim “deadliest years”
- Violent crime $\neq$ deadly (most are non-lethal)

Pattern: All errors bias *toward* overstating crime increases

Group Activity: Peer Review

Exchange Group Activity 3 with another group

Evaluate their proposed measures:

① **Validity assessment:**

- Do the variables proposed have validity?
- Do they capture the intended concept fully?
- Do they avoid capturing other concepts?
- Why or why not?

② **Measurement error assessment:**

- Do these measures suffer from measurement error?
- If so, what kind? (systematic or random)
- What is the likely direction/magnitude?
- How would it affect conclusions?

Time: 15 minutes

Summary: Validity vs. Measurement Error

Concept	Validity	Measurement Error
What it is	Concept-variable mismatch	Variable-measure deviation
The problem	Wrong thing measured	Right thing, wrong numbers
Persists when...	Even with perfect data collection	Even with good concept match
Example	Using violent crime totals to measure "deadliness"	Using incomparable NIBRS coverage across years

Both matter for research quality! Always ask:

- Does this measure capture my concept? (validity)
- Is the measurement process reliable and unbiased? (error)

How to improve validity and reduce error:

① Think carefully about concepts

- Define exactly what you want to measure
- Check if your variable truly captures it

② Understand your data sources

- Who reports? What gets included?
- Has coverage/method changed over time?

③ Use appropriate comparisons

- Rates vs. raw counts (account for denominators)
- Consistent measurement systems across time

④ Be transparent about limitations

- Acknowledge validity issues
- Discuss likely direction of bias